Calculative Foundation: Linear Algebra in Student Performance Analysis

Comprehensive Technical Report & Theoretical Foundations

Project Title: Calculative Foundation (Theory + Practical)
Allocated Duration: 6 Hours | **Domain:** Applied Linear Algebra & Data Science
Author / Candidate: Data Analytics Research Team

1. Dataset Representation

The research dataset records academic examination performances of 5 students across 5 core subjects: `Math`, `Physics`, `Chemistry`, `CS`, and `English`.

Student	Math	Physics	Chemistry	CS	English	Total (L1 Norm)	Magnitude (L2 Norm)
Aarav	85	90	78	92	88	433	193.95
Bhavya	70	75	80	65	72	362	162.28
Chirag	92	88	95	90	85	450	201.59
Diya	60	65	70	58	62	315	141.26
Esha	78	82	84	80	79	403	180.35

2. Part A: Vector Fundamentals & Operations

2.1 Norm-1 (L_1) and Norm-2 (L_2)

```
L1 Norm: ||v||_1 = ∑ |v_i| (Manhattan distance / Aggregate sum)
L2 Norm: ||v||_2 = √(∑ v_i^2) (Euclidean distance / Vector magnitude)
```

- **Aarav:** ||v_A||_1 = 433, ||v_A||_2 = 193.95
- **Bhavya:** ||v_B||_1 = 362, ||v_B||_2 = 162.28

> **Interpretation:** Norm-1 reflects total marks scored across all modules, treating all subjects equally without penalizing outlier marks. Norm-2 represents the true Euclidean vector length, heavily penalizing extreme scores or highlighting students with well-rounded, elevated proficiencies.

2.2 Dot Product and Angle Between Vectors

```
v_A • v_B = ∑ (v_A,i × v_B,i) = 31,256
cos(θ) = (v_A • v_B) / (||v_A||_2 × ||v_B||_2) = 31,256 / (193.9510 × 162.2775) =
```

```
0.9931
θ = arccos(0.9931) = 0.1177 rad = 6.75°
```

> **Interpretation:** The cosine similarity is 0.9931 (angle θ = 6.75°), extremely close to 0°. This proves that Aarav and Bhavya have nearly identical subject profile shapes (balance of marks across domains), even though Aarav achieves higher raw scores overall.

2.3 Cross Product (3D Selected Subjects: Math, Physics, Chemistry)

```
v_A(3D) = [85, 90, 78], v_B(3D) = [70, 75, 80]
v_A × v_B = [ (90×80 - 78×75), -(85×80 - 78×70), (85×75 - 90×70) ]
v_A × v_B = [ 1350, -1340, 75 ]
```

> **Interpretation:** The cross product generates a vector orthogonal (normal) to the 2D subspace spanned by both students in 3D score space. The magnitude of this vector represents twice the area of the triangle formed by their score vectors in 3D.

2.4 Vector Projection

```
proj_{v_B}(v_A) = [(v_A • v_B) / ||v_B||_2^2] × v_B
Scalar factor = 31,256 / 26,334 = 1.1869
Projection Vector = 1.1869 × [70, 75, 80, 65, 72] = [83.08, 89.02, 94.95, 77.15,
85.46]
```

> **Interpretation:** The projection shows the component of Aarav's academic profile aligned directly along Bhavya's score trajectory. The scaling factor of 1.19 indicates Aarav operates at roughly 119% of Bhavya's intensity along that profile.

3. Part B: Matrix Operations

3.1 Matrix Representations & Basic Operations

The student data forms a matrix $A \in \mathbb{R}^{5 imes 5}$.

- **Matrix Addition (A + Bonus):** Adding uniform 5 bonus points across all rows shifts all student vectors uniformly without altering covariance or correlation structures.
- **Transpose (A^T):** Swaps rows and columns, turning student-oriented vectors into subject-oriented vectors in $\mathbb{R}^5$.
- **Gram Matrix ($A^T A$):** Represents pairwise dot products between subjects, capturing uncentered correlations.

3.2 Determinant and Invertibility

```
det(A) = 13,125.0 ≠ 0
```

> **Interpretation:** Since $\det(A) eq 0$, matrix A is **non-singular** and has **full rank (Rank = 5)**. This confirms that all five students possess linearly independent performance profiles, meaning no student's marks can be predicted as a trivial linear combination of other students. Consequently, the unique inverse matrix A^{-1} exists.

4. Part C: Linear Transformations & Geometry

4.1 Line, Plane, and Hyperplane in Dataset Dimensions

Geometric Object	Dimension (k)	Ambient Space (n)	Equation / Mathematical Form	Interpretation in Student Dataset
Line	1D	2D ($ \mathbb{R} ^2$)	w_1 x_1 + w_2 x_2 = b	Linear relationship between two subjects (e.g., Math vs Physics).
Plane	2D	3D ($ \mathbb{R} ^3$)	w_1 x_1 + w_2 x_2 + w_3 x_3 = b	Academic surface across 3 subjects (e.g., STEM trifecta: Math, Physics, Chem).
Hyperplane	4D	5D ($ \mathbb{R} ^5$)	w^T x = ∑ _{i=1}^5 w_i x_i = b	Decision boundary in the full 5-subject space dividing passing/failing or admission tiers.

4.2 Dimensionality Progression ($2D o 3D o N-D$)

As dimensions scale, every hyperplane in $\mathbb{R}^n$ always has dimension $n-1$. In machine learning (e.g., Support Vector Machines, Linear Regression), hyperplanes serve as global linear decision boundaries dividing the feature space into distinct decision regions.

5. Part D: Eigenvalues, Eigenvectors & Covariance Decomposition

5.1 Covariance Matrix (Σ)

Calculated across the 5 subjects to capture how student scores co-vary.

5.2 Eigenvalue Decomposition

$$\Sigma v = \lambda v$$

Principal Component	Eigenvalue (λ)	Variance Explained (%)	Cumulative Variance (%)
PC 1	632.1655	**92.69%**	92.69%
PC 2	44.8511	**6.58%**	99.27%
PC 3	4.0915	0.60%	99.87%
PC 4	0.8919	0.13%	100.00%
PC 5	0.0000	0.00%	100.00%

Key Data Science Takeaway:

- **PC1 (92.69% of variance):** Demonstrates that student performance is overwhelmingly dominated by a single latent factor: general cognitive/academic drive. When a student performs well in Math, they tend to perform proportionally well across all subjects.
- **PC2 (6.58% of variance):** Captures STEM vs. Humanities divergence (students who excel more in Chemistry/CS versus English).
- **Dimensionality Reduction:** Retaining just **2 Principal Components** preserves **99.27%** of the total information, compressing a 5D problem into a 2D plane with virtually zero information loss.